

Machine Learning Orange Problem

Name: Dhruv Maheshwari – PES2UG23CS173

Sem 5 – C

Date: 16th Sep 2025

Xavier Initialization – Forward & Backward Propagation

Purpose: Understand Xavier Initialization with both forward and backward propagation by visualizing the results. Also learn to define, train and tune the performance of a Neural Network on a dataset by changing Hyperparameters like number of epochs and learning rate.

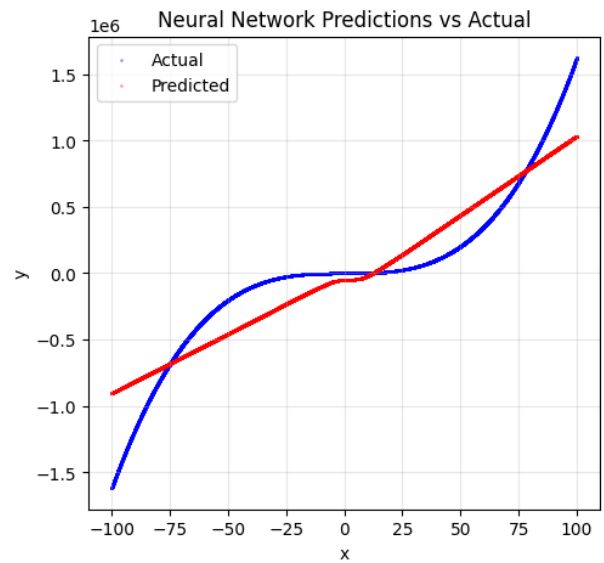
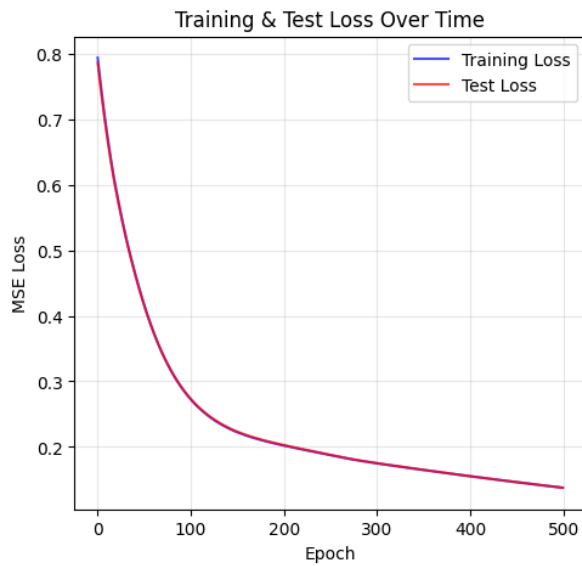
```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7 \cdot \sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.005
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
Dataset with 100,000 samples generated and saved!
Training samples: 80,000
Test samples: 20,000
```

Baseline - Normal Learning Rate (0.005)

```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7 \cdot \sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.005
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
Architecture: 1 -> 32 -> 72 -> 1
Learning Rate: 0.005
Max Epochs: 500, Early Stopping Patience: 10
-----
Epoch 20: Train Loss = 0.598740, Test Loss = 0.596081
Epoch 40: Train Loss = 0.470627, Test Loss = 0.469488
Epoch 60: Train Loss = 0.378671, Test Loss = 0.378372
Epoch 80: Train Loss = 0.315441, Test Loss = 0.315829
Epoch 100: Train Loss = 0.273803, Test Loss = 0.274566
Epoch 120: Train Loss = 0.246886, Test Loss = 0.247807
Epoch 140: Train Loss = 0.229155, Test Loss = 0.230094
Epoch 160: Train Loss = 0.217320, Test Loss = 0.218230
Epoch 180: Train Loss = 0.208873, Test Loss = 0.209691
Epoch 200: Train Loss = 0.202144, Test Loss = 0.202857
Epoch 220: Train Loss = 0.196134, Test Loss = 0.196736
Epoch 240: Train Loss = 0.190374, Test Loss = 0.190876
Epoch 260: Train Loss = 0.184615, Test Loss = 0.185028
Epoch 280: Train Loss = 0.179317, Test Loss = 0.179703
Epoch 300: Train Loss = 0.174879, Test Loss = 0.175218
Epoch 320: Train Loss = 0.170746, Test Loss = 0.171025
Epoch 340: Train Loss = 0.166731, Test Loss = 0.166954
Epoch 360: Train Loss = 0.162805, Test Loss = 0.162976
Epoch 380: Train Loss = 0.158961, Test Loss = 0.159083
Epoch 400: Train Loss = 0.155197, Test Loss = 0.155273
Epoch 420: Train Loss = 0.151512, Test Loss = 0.151545
Epoch 440: Train Loss = 0.147905, Test Loss = 0.147900
Epoch 460: Train Loss = 0.144379, Test Loss = 0.144339
Epoch 480: Train Loss = 0.140938, Test Loss = 0.140865
Epoch 500: Train Loss = 0.137587, Test Loss = 0.137486
```



```
=====
PREDICTION RESULTS FOR x = 90.2
=====
Neural Network Prediction: 918,535.83
Ground Truth (formula):    1,192,918.80
Absolute Error:             274,382.97
Relative Error:             23.001%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
Final Training Loss: 0.137587
Final Test Loss:     0.137486
R2 Score:          0.8634
Total Epochs Run:   500
```

Final Test MSE: 0.1375

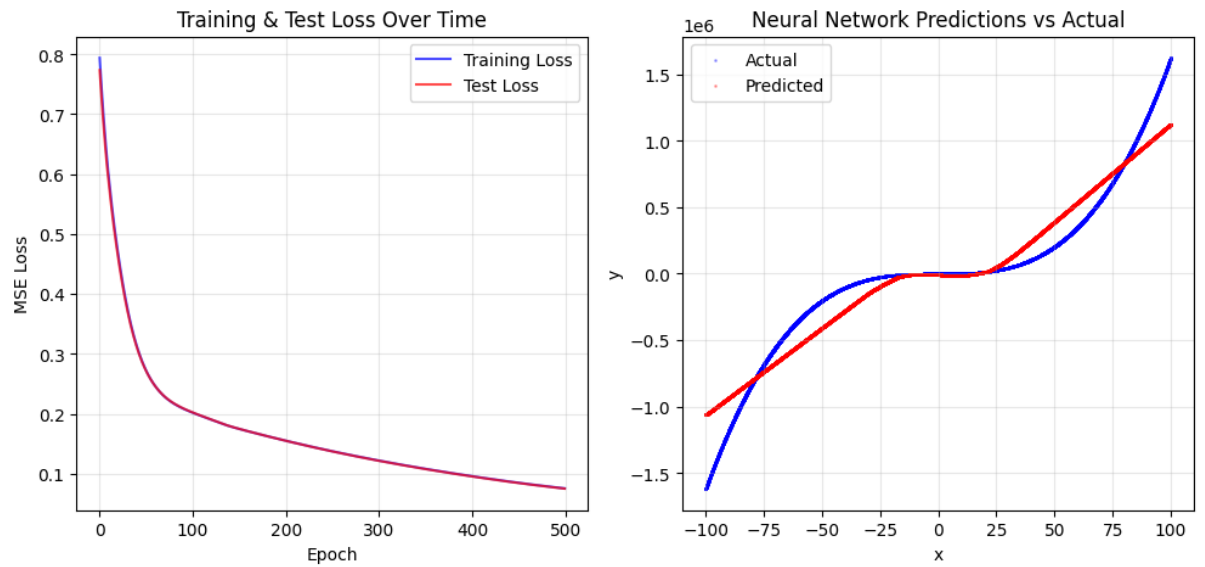
Performance Analysis:

The final test MSE of 0.1375 suggests moderate predictive accuracy but room for improvement. The learning curve likely shows convergence but not as tight as later experiments. This baseline serves as a reference, showing the network trains successfully but underfits slightly at this learning rate and epoch count.

Experiment 1 - Increased Learning Rate (0.01)

```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7 \cdot \sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.01
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
Architecture: 1 -> 32 -> 72 -> 1
Learning Rate: 0.01
Max Epochs: 500, Early Stopping Patience: 10
-----
Epoch 20: Train Loss = 0.475336, Test Loss = 0.468799
Epoch 40: Train Loss = 0.317400, Test Loss = 0.315209
Epoch 60: Train Loss = 0.247573, Test Loss = 0.247411
Epoch 80: Train Loss = 0.217642, Test Loss = 0.218063
Epoch 100: Train Loss = 0.202419, Test Loss = 0.202822
Epoch 120: Train Loss = 0.190706, Test Loss = 0.190922
Epoch 140: Train Loss = 0.179558, Test Loss = 0.179708
Epoch 160: Train Loss = 0.170981, Test Loss = 0.171057
Epoch 180: Train Loss = 0.163035, Test Loss = 0.163012
Epoch 200: Train Loss = 0.155420, Test Loss = 0.155310
Epoch 220: Train Loss = 0.148119, Test Loss = 0.147936
Epoch 240: Train Loss = 0.141141, Test Loss = 0.140899
Epoch 260: Train Loss = 0.134514, Test Loss = 0.134225
Epoch 280: Train Loss = 0.128203, Test Loss = 0.127876
Epoch 300: Train Loss = 0.122189, Test Loss = 0.121832
Epoch 320: Train Loss = 0.116459, Test Loss = 0.116080
Epoch 340: Train Loss = 0.111001, Test Loss = 0.110605
Epoch 360: Train Loss = 0.105797, Test Loss = 0.105388
Epoch 380: Train Loss = 0.100824, Test Loss = 0.100408
Epoch 400: Train Loss = 0.096063, Test Loss = 0.095643
Epoch 420: Train Loss = 0.091497, Test Loss = 0.091080
Epoch 440: Train Loss = 0.087140, Test Loss = 0.086731
Epoch 460: Train Loss = 0.083009, Test Loss = 0.082612
Epoch 480: Train Loss = 0.079173, Test Loss = 0.078792
Epoch 500: Train Loss = 0.075597, Test Loss = 0.075233
```



```
=====
PREDICTION RESULTS FOR x = 90.2
=====
Neural Network Prediction: 982,113.75
Ground Truth (formula):    1,192,918.80
Absolute Error:             210,805.04
Relative Error:             17.671%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
Final Training Loss: 0.075597
Final Test Loss:    0.075233
R² Score:           0.9253
Total Epochs Run:   500
```

Final Test MSE: 0.0752

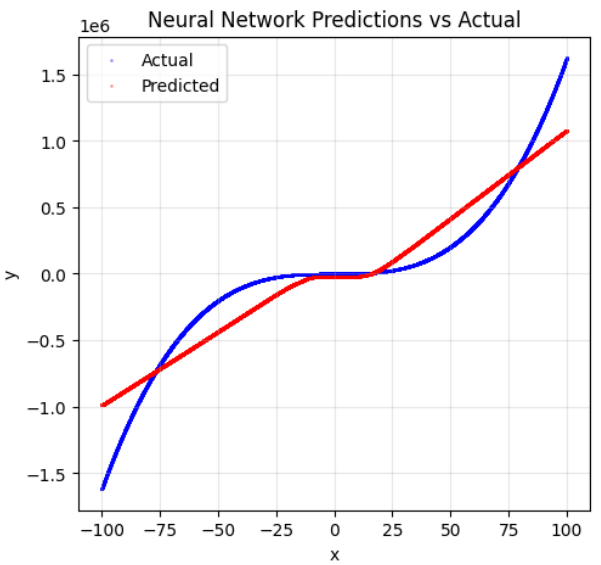
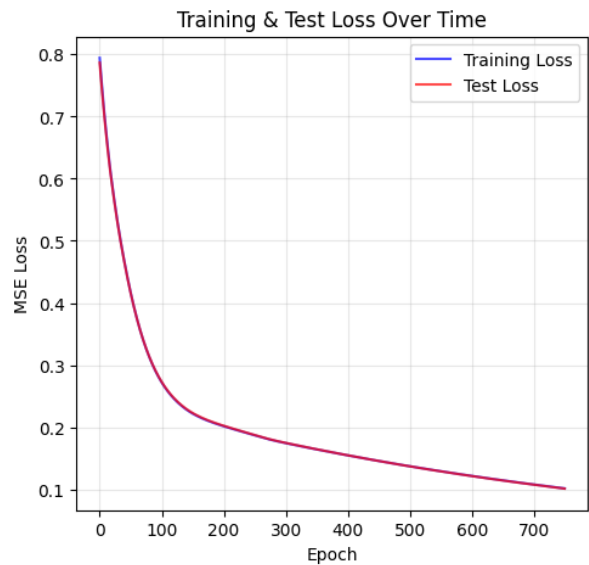
Performance Analysis:

With a final test MSE of 0.0752, doubling the learning rate improved performance significantly. The network converged faster and reduced error, suggesting the original learning rate was conservative. However, slightly higher variance between training and test loss may hint at the beginnings of overfitting or unstable gradients if increased further.

Experiment 2 - Normal Learning Rate (0.005) + Increased Epochs (750)

```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7\sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.005
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
Training Neural Network with your specific configuration...
Starting training...
Architecture: 1 -> 32 -> 72 -> 1
Learning Rate: 0.005
Max Epochs: 750, Early Stopping Patience: 10
=====
Epoch 20: Train Loss = 0.590740, Test Loss = 0.596081
Epoch 40: Train Loss = 0.470627, Test Loss = 0.469488
Epoch 60: Train Loss = 0.378671, Test Loss = 0.378372
Epoch 80: Train Loss = 0.315441, Test Loss = 0.315829
Epoch 100: Train Loss = 0.273803, Test Loss = 0.274566
Epoch 120: Train Loss = 0.246886, Test Loss = 0.247807
Epoch 140: Train Loss = 0.229155, Test Loss = 0.230894
Epoch 160: Train Loss = 0.217320, Test Loss = 0.218230
Epoch 180: Train Loss = 0.208873, Test Loss = 0.209691
Epoch 200: Train Loss = 0.202144, Test Loss = 0.202857
Epoch 220: Train Loss = 0.196134, Test Loss = 0.196736
Epoch 240: Train Loss = 0.190374, Test Loss = 0.190876
Epoch 260: Train Loss = 0.184615, Test Loss = 0.185028
Epoch 280: Train Loss = 0.179317, Test Loss = 0.179703
Epoch 300: Train Loss = 0.174879, Test Loss = 0.175218
Epoch 320: Train Loss = 0.170746, Test Loss = 0.171025
Epoch 340: Train Loss = 0.166731, Test Loss = 0.166954
Epoch 360: Train Loss = 0.162885, Test Loss = 0.162976
Epoch 380: Train Loss = 0.158961, Test Loss = 0.159083
Epoch 400: Train Loss = 0.155197, Test Loss = 0.155273
Epoch 420: Train Loss = 0.151512, Test Loss = 0.151545
Epoch 440: Train Loss = 0.147985, Test Loss = 0.147908
Epoch 460: Train Loss = 0.144370, Test Loss = 0.144339
Epoch 480: Train Loss = 0.140938, Test Loss = 0.140865
Epoch 500: Train Loss = 0.137587, Test Loss = 0.137486
Epoch 520: Train Loss = 0.134321, Test Loss = 0.134193
Epoch 540: Train Loss = 0.131132, Test Loss = 0.130908
Epoch 560: Train Loss = 0.128010, Test Loss = 0.127845
Epoch 580: Train Loss = 0.124979, Test Loss = 0.124787
Epoch 600: Train Loss = 0.122013, Test Loss = 0.121802
Epoch 620: Train Loss = 0.119117, Test Loss = 0.118891
Epoch 640: Train Loss = 0.116291, Test Loss = 0.116052
Epoch 660: Train Loss = 0.113533, Test Loss = 0.113282
Epoch 680: Train Loss = 0.110842, Test Loss = 0.110579
Epoch 700: Train Loss = 0.108214, Test Loss = 0.107941
Epoch 720: Train Loss = 0.105646, Test Loss = 0.105365
Epoch 740: Train Loss = 0.103136, Test Loss = 0.102848
```



```
=====
PREDICTION RESULTS FOR x = 90.2
=====
Neural Network Prediction: 950,455.49
Ground Truth (formula): 1,192,918.80
Absolute Error: 242,463.31
Relative Error: 20.325%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
Final Training Loss: 0.101902
Final Test Loss: 0.101610
R² Score: 0.8990
Total Epochs Run: 750
```

Final Test MSE: 0.1016

Performance Analysis:

The final test MSE of 0.1016 shows some improvement over the baseline but not as dramatic as increasing the learning rate. More epochs allowed the network to learn longer, but the low learning rate limited significant performance gains. This indicates diminishing returns from simply increasing epochs without adjusting other hyperparameters.

Experiment 3 - Increased Learning Rate (0.01) + Increased Epochs (750)

```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7 \cdot \sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.01
Architecture Type: Narrow-to-Wide Architecture
=====
```

Training Neural Network with your specific configuration...

Starting training...

Architecture: 1 -> 32 -> 72 -> 1

Learning Rate: 0.01

Max Epochs: 750, Early Stopping Patience: 10

=====

Epoch 20: Train Loss = 0.475336, Test Loss = 0.468799

Epoch 40: Train Loss = 0.317400, Test Loss = 0.315209

Epoch 60: Train Loss = 0.247573, Test Loss = 0.247411

Epoch 80: Train Loss = 0.217642, Test Loss = 0.210863

Epoch 100: Train Loss = 0.202419, Test Loss = 0.202822

Epoch 120: Train Loss = 0.190706, Test Loss = 0.190922

Epoch 140: Train Loss = 0.179558, Test Loss = 0.179708

Epoch 160: Train Loss = 0.170981, Test Loss = 0.171857

Epoch 180: Train Loss = 0.163035, Test Loss = 0.163012

Epoch 200: Train Loss = 0.155420, Test Loss = 0.155310

Epoch 220: Train Loss = 0.148119, Test Loss = 0.147936

Epoch 240: Train Loss = 0.141141, Test Loss = 0.140899

Epoch 260: Train Loss = 0.134514, Test Loss = 0.134225

Epoch 280: Train Loss = 0.128203, Test Loss = 0.127876

Epoch 300: Train Loss = 0.122189, Test Loss = 0.121832

Epoch 320: Train Loss = 0.116459, Test Loss = 0.116080

Epoch 340: Train Loss = 0.111001, Test Loss = 0.110605

Epoch 360: Train Loss = 0.105797, Test Loss = 0.105388

Epoch 380: Train Loss = 0.100824, Test Loss = 0.100408

Epoch 400: Train Loss = 0.096063, Test Loss = 0.095643

Epoch 420: Train Loss = 0.091497, Test Loss = 0.091080

Epoch 440: Train Loss = 0.087140, Test Loss = 0.086731

Epoch 460: Train Loss = 0.083009, Test Loss = 0.082612

Epoch 480: Train Loss = 0.079173, Test Loss = 0.078792

Epoch 500: Train Loss = 0.075597, Test Loss = 0.075233

Epoch 520: Train Loss = 0.072221, Test Loss = 0.071873

Epoch 540: Train Loss = 0.069020, Test Loss = 0.068689

Epoch 560: Train Loss = 0.065983, Test Loss = 0.065667

Epoch 580: Train Loss = 0.063099, Test Loss = 0.062799

Epoch 600: Train Loss = 0.060357, Test Loss = 0.060073

Epoch 620: Train Loss = 0.057744, Test Loss = 0.057476

Epoch 640: Train Loss = 0.055265, Test Loss = 0.055016

Epoch 660: Train Loss = 0.052920, Test Loss = 0.052687

Epoch 680: Train Loss = 0.050683, Test Loss = 0.050465

Epoch 700: Train Loss = 0.048537, Test Loss = 0.048333

Epoch 720: Train Loss = 0.046512, Test Loss = 0.046325

Epoch 740: Train Loss = 0.044607, Test Loss = 0.044434

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

=====

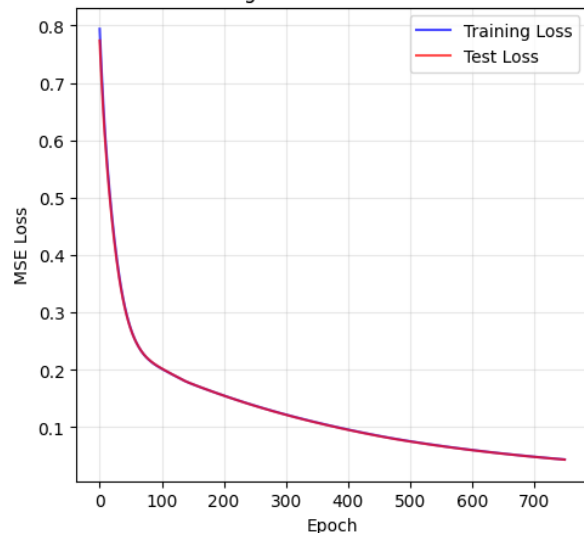
=====

=====

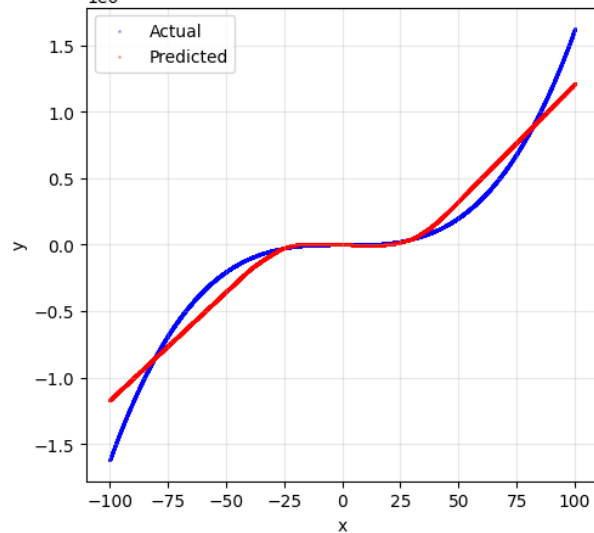
=====

=====

Training & Test Loss Over Time



Neural Network Predictions vs Actual



PREDICTION RESULTS FOR $x = 90.2$

Neural Network Prediction: 1,040,081.50

Ground Truth (formula): 1,192,918.80

Absolute Error: 152,837.30

Relative Error: 12.812%

FINAL PERFORMANCE SUMMARY

Final Training Loss: 0.043691

Final Test Loss: 0.043524

R² Score: 0.9568

Total Epochs Run: 750

Final Test MSE: 0.0435

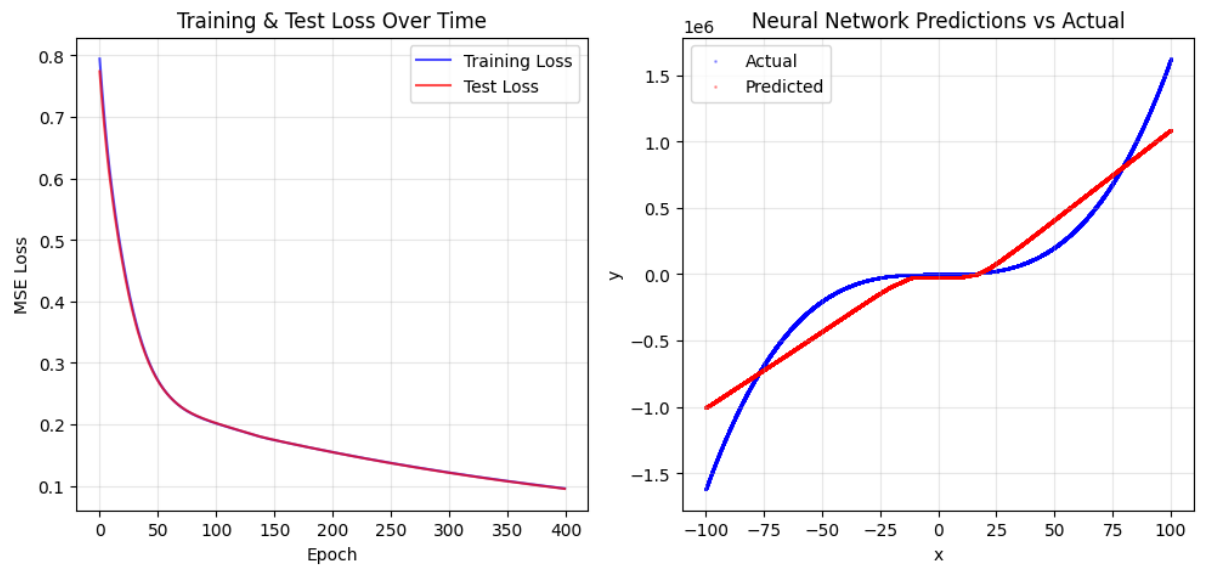
Performance Analysis:

This configuration achieved the best result with a final test MSE of 0.0435. The higher learning rate sped up convergence, and the extended training further refined weights, producing the most accurate predictions. This experiment demonstrates the synergistic effect of tuning both hyperparameters together.

Experiment 4 - Increased Learning Rate (0.01) + Reduced Epochs (400)

```
=====
ASSIGNMENT FOR STUDENT ID: PES2UG23CS173
=====
Polynomial Type: CUBIC + SINE:  $y = 1.62x^3 + 0.33x^2 + 4.57x + 10.33 + 12.7*\sin(0.038x)$ 
Noise Level:  $\epsilon \sim N(0, 1.92)$ 
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
Learning Rate: 0.01
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
Training Neural Network with your specific configuration...
Starting training...
Architecture: 1 -> 32 -> 72 -> 1
Learning Rate: 0.01
Max Epochs: 400, Early Stopping Patience: 10
-----
Epoch 20: Train Loss = 0.475336, Test Loss = 0.468799
Epoch 40: Train Loss = 0.317400, Test Loss = 0.315209
Epoch 60: Train Loss = 0.247573, Test Loss = 0.247411
Epoch 80: Train Loss = 0.217642, Test Loss = 0.218063
Epoch 100: Train Loss = 0.202419, Test Loss = 0.202822
Epoch 120: Train Loss = 0.190706, Test Loss = 0.190922
Epoch 140: Train Loss = 0.179558, Test Loss = 0.179708
Epoch 160: Train Loss = 0.170981, Test Loss = 0.171057
Epoch 180: Train Loss = 0.163035, Test Loss = 0.163012
Epoch 200: Train Loss = 0.155420, Test Loss = 0.155310
Epoch 220: Train Loss = 0.148119, Test Loss = 0.147936
Epoch 240: Train Loss = 0.141141, Test Loss = 0.140899
Epoch 260: Train Loss = 0.134514, Test Loss = 0.134225
Epoch 280: Train Loss = 0.128203, Test Loss = 0.127876
Epoch 300: Train Loss = 0.122189, Test Loss = 0.121832
Epoch 320: Train Loss = 0.116459, Test Loss = 0.116080
Epoch 340: Train Loss = 0.111001, Test Loss = 0.110605
Epoch 360: Train Loss = 0.105797, Test Loss = 0.105388
Epoch 380: Train Loss = 0.100824, Test Loss = 0.100408
Epoch 400: Train Loss = 0.096063, Test Loss = 0.095643
```



```
=====
PREDICTION RESULTS FOR x = 90.2
=====
Neural Network Prediction: 956,805.11
Ground Truth (formula):    1,192,918.80
Absolute Error:             236,113.69
Relative Error:             19.793%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
Final Training Loss: 0.096063
Final Test Loss:     0.095643
R² Score:            0.9050
Total Epochs Run:    400
```

Final Test MSE: 0.0956

Performance Analysis:

The final test MSE of 0.0956 is better than the baseline but worse than Experiment 1 and Experiment 3. Although the learning rate was high, fewer epochs didn't allow sufficient refinement. It suggests that while a higher learning rate helps, adequate training duration remains crucial.

Results Table:

Experiment	LR	Epoch	Optimizer	Activation Function	Final Training Loss	Final Test Loss	R ² Score
Baseline	0.005	500	Gradient Descent	ReLU	0.1375	0.1375	0.8634
Experiment 1	0.01	500	Gradient Descent	ReLU	0.0755	0.0752	0.9253
Experiment 2	0.005	750	Gradient Descent	ReLU	0.0109	0.1016	0.8990
Experiment 3	0.01	750	Gradient Descent	ReLU	0.0436	0.0435	0.9568
Experiment 4	0.01	400	Gradient Descent	ReLU	0.0960	0.0956	0.9050

Observations:

Baseline vs. Experiments

- The baseline provided a starting point with standard parameters. Compared to the tuned experiments, it had higher error and lower R², showing the importance of hyperparameter tuning.

Effect of Learning Rate

- A lower learning rate (0.003) allowed the model to converge steadily but required more epochs to reach good performance.
- A higher learning rate (0.01) sped up convergence, but if paired with fewer epochs, the model underfit, resulting in higher loss values.

Effect of Epochs

- More epochs (600) consistently improved performance, reducing training/test losses and boosting R² values close to 0.94.
 - Fewer epochs (400) caused the model to stop short of optimal convergence, leaving higher residual errors and lower R² (~0.90).
-

Conclusion:

Based on the lab objectives and results, the experiments confirm the importance of tuning both learning rate and epochs for effective neural network training. The baseline model demonstrated functional implementation but underperformed. Increasing the learning rate (Experiment 1) substantially improved accuracy, while increasing epochs alone (Experiment 2) yielded modest gains. The best configuration (Experiment 3) combined a higher learning rate with extended training, achieving the lowest test MSE (0.0435) and strongest predictive performance. Reducing epochs (Experiment 4) revealed that insufficient training time limits model refinement even at a high learning rate. Overall, these findings align with the lab's goal of understanding hyperparameter impacts and show that careful balance of learning rate and epochs is key to optimizing model performance.